Patnam Shariq Faraz Muhammed

ee24btech11049@iith.ac.in | pshariqfaraz@gmail.com | +91 7893293604 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.32
Narayana Junior College	Class XII (TSBIE)	2024	98.40%
Narayana School	Class X (SSC)	2022	10.00%

SKILLS

Programming Languages: C, RISC-V Assembly, Python

Hardware Description Languages: Verilog

Tools: MATLAB, Arduino IDE, Git/GitHub, LTspice, Altium Designer, LaTeX, Linux, Quartus Prime Lite

Frameworks & Libraries: NumPy, Matplotlib, pandas, scikit-learn, OpenCV, ROS 2, Nav2

RELEVANT COURSES

Mathematics: Mathematics for Electrical Engineering, Vector Calculus, Matrix Theory, Differential Equations and Transform Techniques, Probability, Random Processes, Numerical Analysis, Introduction to Statistics

Electrical Engineering: Scientific Programming for Electrical Engineering, Digital Systems, Signal Processing

Computer Science & AI: Introduction to Programming, Data Structures and Applications, Machine Learning

Laboratories: Digital Systems Lab, Digital Signal Processing Lab

PROJECTS

- **e-Yantra Robotics Competition (eYRC) | Krishi Cobot Theme | Github:**
- Implemented autonomous navigation using **LiDAR** data, **ROS2 Odometry**, and **PID tuning** for ebot.
- Controlled the arm using **TF transforms** and **PID tuning** for accurate waypoint navigation. Implemented **ArUco marker detection** for fertilizer pickup and **OpenCV**-based color detection for real-time fruit detection and classification.
- Tuned the **UR5 arm's PID controllers** to minimize fertilizer transfer time onto the eBot and optimize autonomous pick-and-place fruit segregation. Validated the complete system in **Gazebo** and **RViz**.

- **Inter IIT Tech Meet 14.0 | Drona Aviation Problem Statement | Github**:
- Designed and tuned cascaded **PID controllers** for altitude, position, velocity, and attitude control to achieve **GPS-denied stabilization** of a nano-drone.
- Developed and refined sensor fusion pipelines using **Kalman Filters**, **Extended Error-State Kalman Filter (ESKF)**, and **complementary filtering** for robust state estimation and drift reduction.
- Debugged and optimized the multi-stage control architecture to improve hover stability, micro-movement accuracy, and recovery from state estimation errors.

MINI PROJECTS

- **Medical Insurance Cost Prediction | Python | GitHub:**
- Developed a **regression model** to predict medical insurance charges using demographic and lifestyle features.
- Performed **exploratory data analysis, feature engineering, and categorical feature encoding** to improve model performance.
- Built a reproducible machine learning workflow covering **data preprocessing, model training, and evaluation** using Scikit-learn.

- **MNIST Neural Network from Scratch | Python | GitHub:**
- Implemented a **feedforward neural network from scratch** for handwritten digit classification without using deep learning frameworks.
- Built the complete training pipeline including **forward propagation, backpropagation, gradient descent, and parameter updates**.
- Implemented matrix-based computations using **NumPy** and evaluated classification performance on the MNIST dataset.

- **Scientific Calculator using Numerical Methods | Arduino UNO | Github:**
- Designed an infix-to-postfix expression parser using the **Shunting Yard algorithm** supporting operator precedence for scientific functions.
- Implemented **RK4** for trigonometric functions, **Euler's method** for exponentiation, and **Trapezoidal integration** for logarithms.
- Built a **16×2 LCD-based calculator** with a resistor-ladder keypad interface to maximize input functionality on the Arduino Uno.

- **Eigenvalue Solver for Complex Matrices | C | GitHub:**
- Implemented an iterative **QR algorithm** using **Gram-Schmidt orthogonalization** to compute eigenvalues of complex-valued matrices.
- Developed support for **complex matrix operations** and iterative QR decomposition using native C complex arithmetic.
- Implemented eigenvalue extraction from **2×2 matrix blocks** by solving the characteristic polynomial to compute complex-conjugate eigenvalue pairs.

POSITIONS OF RESPONSIBILITY

- **Teaching Assistant, EE1080: Probability**
  Conducted practice sessions and designed and graded programming assignments.

- **Core Member, Electronics and Embedded Systems** - Robotix, IIT Hyderabad

- **Senior Core, Programming and ML** - Robotix, IIT Hyderabad